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RESEARCH ARTICLE

Machine Learning Framework for Early Detection of Chronic Kidney Disease Stages Using Optimized Estimated Glomerular Filtration Rate

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ABSTRACT Chronic Kidney Disease (CKD) is a progressive condition that requires accurate diagnosis and staging for effective clinical management. Conventional CKD diagnosis relies on estimated Glomerular Filtration Rate (eGFR), a measure of kidney function derived from serum biomarkers such as serum creatinine (SCr) and cystatin C (SCysC). However, eGFR calculations may be inaccurate when applied to diverse patient populations. This study proposes a machine learning (ML) system that integrates regression-based eGFR estimation, metaheuristic optimization using the Grey Wolf Optimizer (GWO), and multi-class classification with various ML models to enhance CKD staging and classification. The model estimates eGFR using three established CKD Epidemiology Collaboration (CKD-EPI) equations incorporating SCr, SCysC, and their combined values. Regression models assess predictive performance, specifically Linear Regression (LR) and Support Vector Regression (SVR). SVR demonstrates superior performance compared to LR for CKD-EPI_{SCr-SCysC} achieved a root mean squared error (RMSE) of 3.03, a mean absolute percentage error (MAPE) of 2.97%, and a coefficient of determination (R^2) score of 0.97. The application of GWO for hyperparameter tuning has resulted in a 37.3% reduction in root mean square error (RMSE), a 37.4% drop in mean absolute percentage error (MAPE), and a 2.06% improvement in R^2 to improve the precision of prediction. Once the model fine-tunes the eGFR estimations, it feeds them into various algorithms for CKD stage classification, including Support Vector Machine (SVM), Decision Tree (DT), Random Forest (RF), and Extreme Gradient Boosting (XGBoost). Among these, XGBoost achieves the highest classification accuracy of 97.76%, along with an F1-score of 97.45%, demonstrating its effectiveness in CKD staging. Shapley Additive Explanations (SHAP) provide global and local feature importance insights, enhancing clinical decision-making and model transparency. Future research will validate the model using more extensive and more diverse datasets. Additionally, it will incorporate extra clinical parameters, including biomarkers and genetic data, to enhance the precision of CKD risk prediction. This research enhances AI-driven nephrology by providing a scalable, interpretable, and highly accurate solution for diagnosing and managing CKD.

INDEX TERMS Chronic kidney diseases, CKD-EPI equation, cystatin C, glomerular filtration rate, serum creatinine, machine learning, explainable AI.

I. INTRODUCTION

Chronic kidney disease (CKD) is a leading cause of mortality and impacts an estimated 850 million individuals globally [1]. It is expected to become the world's fifth most common cause

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of death by 2040 [2]. Early identification and appropriate care of declining kidney function can dramatically reduce the progression of CKD to more severe stages, such as end-stage kidney disease (ESKD) [3], [4], [5]. Detecting CKD early is essential for managing the disease and improving patient outcomes, yet early-stage CKD often lacks noticeable symptoms [6]. CKD is classified into five stages based on the glomerular filtration rate (GFR), a measure of kidney function. GFR indicates how effectively the kidneys filter waste from the blood. Accurate measurement of GFR is crucial for the early detection, staging, and management of CKD. A decline in GFR may indicate the presence of CKD or acute kidney injury (AKI) [7], [8]. According to the Kidney Disease Improving Global Outcomes (KDIGO) guidelines, determining the GFR is critical in evaluating individuals with CKD [9].

GFR can be measured using inulin or by assessing the clearance of external filtration markers such as technetium Tc-99m diethylenetriamine pentaacetic acid, chromium 51-ethylenediamine tetraacetic acid, or iothexol [10]. However, these methods are not commonly used in clinical practice due to their complexity and high cost. Consequently, the estimated GFR (eGFR), derived from eGFR equations, has become a more practical tool for evaluating kidney function and diagnosing CKD [11]. While eGFR equations are essential for evaluating kidney disease at individual and population levels, they have notable limitations. These equations often produce inaccurate predictions at the individual level, imprecise population forecasts for CKD-related conditions such as diabetes, and bias at GFR levels above 60 mL/min/1.73m². Moreover, the use of race adjustments in serum creatinine (SCr)-based GFR estimation equations for Black individuals is a significant issue [12], [13], [14], [15], [16].

Traditional SCr-based equations have long faced criticism for their effectiveness in detecting early-stage CKD. These equations underestimate disease prevalence, as SCr levels often remain within the reference range until about 50% of glomerular function is lost. This period is commonly referred to as the “creatinine-insensitive zone,” where eGFR typically falls between 60–80 mL/min/1.73m². In comparison, estimates based on serum cystatin C (SCysC) provide more accurate early detection [17], [18]. Recent research has also shown that a significant portion of the global population remains undetected at more advanced CKD stages when using SCr-based equations, such as Modification of Diet in Renal Disease (MDRD), Cockcroft-Gault (CG), and CKD EPI, which tend to overestimate GFR in patients with GFR ≤ 60 mL/min/1.73m² [17], [19].

The use of a CysC-based equation is especially critical for detecting declining kidney function early, particularly among high-risk patient groups like those with diabetes and hypertension. These patients may mistakenly believe they don't have CKD due to its asymptomatic nature in the initial stages. Previous studies have shown that CysC is an

effective biomarker for early disease diagnosis, especially in patients with hypertensive nephropathy [17]. Unfortunately, CKD is often diagnosed at advanced stages, frequently at CKD stage 3, leaving dialysis as the sole viable option. Timely clinical intervention is essential to mitigate such outcomes [17].

To estimate the GFR, clinicians often rely on SCr and serum SCysC levels [20]. These biomarkers have long been identified as critical indicators of kidney function, with numerous studies demonstrating their utility in estimating GFR and diagnosing CKD [21], [22], [23], [24], [25]. However, while SCr has been the primary marker used in many clinical settings, its ability to accurately reflect kidney function is limited due to various factors, including age, race, diet, muscle mass, drugs, and renal tubular secretion [26]. Additionally, SCr is susceptible to external influences, making it a less reliable sole marker for kidney health [27]. In contrast, SCysC, a protein produced by all nucleated cells and excreted by the kidneys, has been proposed as an alternative marker due to its relatively stable production, unaffected by muscle mass and diet. However, while SCysC has its advantages, certain conditions—such as smoking, inflammation, obesity, and the use of glucocorticoids—can influence its levels, potentially affecting its diagnostic accuracy. Combining SCr and SCysC in eGFR equations has enhanced the prediction of CKD and related outcomes, including heart disease, mortality, and end-stage kidney disease (ESKD) [22], [28].

Researchers have developed several formulas to calculate eGFR, an essential indicator of kidney function. Each equation has unique strengths and limitations, contributing to ongoing discussions regarding their precision and effectiveness. The most widely used equations for estimating GFR include the Modification of Diet in Renal Disease (MDRD) and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) formulas, which incorporate SCr and demographic data such as age, gender, and ethnicity. The CKD-EPI equation has been recognized for superior performance when the GFR exceeds 90 mL/min. Despite their popularity, these equations have limitations, particularly regarding precision at higher GFR levels and their reliance on race for SCr-based GFR estimation. Researchers have proposed incorporating renal biomarkers, like SCysC, to enhance the sensitivity and specificity of eGFR equations. However, comprehensive studies comparing the reliability and correlation of these equations with accurate disease measurement remain scarce.

Machine learning (ML) techniques have been extensively explored for estimating GFR and diagnosing or staging CKD. These models have demonstrated the potential to enhance kidney function prediction and improve early detection. For instance, Nakano et al. [29] applied random forest models to predict GFR and compared the results with the European Kidney Function Consortium (EKFC) equation. Their study revealed that ML performed comparably to EKFC, with

a slight advantage in younger patients, underscoring the promise of ML in refining kidney function assessments. Similarly, Dastidar et al. [17] examined various equations for eGFR using both SCr and SCysC. Their findings indicated that integrating SCysC with SCr-based equations enhanced early CKD detection, offering improved sensitivity and specificity.

ML's capability to analyze vast datasets enables it to identify hidden patterns and risk factors often overlooked by conventional methods. This advantage facilitates the early and accurate detection of CKD stages, allowing healthcare providers to adopt proactive measures to slow disease progression. Despite numerous ML-based CKD prediction models, most focus on binary classification [30], [31], [32]. Our prior work also developed explainable ML models for binary CKD prediction [33]. However, this binary approach oversimplifies CKD, which progresses through multiple stages with varying degrees of dysfunction.

Several studies have explored ML techniques for CKD stage classification. Rady and Anwar [34] compared multiple algorithms for CKD stage prediction, including multilayer perceptron, probabilistic neural network (PNN), radial basis function (RBF), and support vector machine (SVM). Among these, PNN exhibited the highest performance, achieving a 94% F-measure score in severity classification. However, they trained and tested their model on a relatively small dataset, potentially limiting its generalizability. Additionally, Jeong et al. [35] assessed statistical and ML models for CKD stage classification using an imbalanced dataset from Korea's national health insurance service. Their evaluation included multinomial logistic regression, ordinal logistic regression, random forest (RF), and an autoencoder (AE). While RF and AE outperformed the regression models with over 99% accuracy, the study highlighted a crucial limitation—accuracy alone was insufficient when dealing with highly imbalanced data, as the models exhibited poor F-measure scores across different CKD stages. These findings underscore the need for robust ML models that achieve high accuracy and maintain balanced performance across all CKD stages, mainly when working with imbalanced datasets.

The review above indicates that numerous studies have explored ML techniques for CKD prediction, with factors like dataset size, quality, and timing influencing model performance. However, many models face challenges in clinical adoption due to their opaque nature [36], [37]. Understanding this “opaque” is essential, as it enables physicians to comprehend the inner workings of ML models. To address this challenge, explainable artificial intelligence (XAI) has been developed to clarify how ML models make decisions. Explainable ML focuses on enhancing the transparency and credibility of decision-making in these “opaque” models. Among various XAI techniques, Shapley Additive exPlanations (SHAP) are crucial in providing insights into CKD prediction models [38]. Nevertheless, research on the reliability and robustness of these explanations in the context of CKD prediction remains limited.

This study aims to develop ML-based predictive models that accurately classify CKD stages using clinical biomarkers and demographic factors. To achieve this, various CKD-EPI equations—incorporating SCr, SCysC, Age, and Sex—will be evaluated to determine the most reliable approach for eGFR estimation. The study will further compare ML-based CKD classification with CKD-EPI-based staging, assessing improvements in predictive accuracy, reliability, and clinical applicability. Additionally, feature importance analysis will be conducted to interpret model predictions and identify key clinical factors influencing CKD classification. This research seeks to enhance clinical decision-making, improve CKD detection, and improve risk stratification by integrating clinically established CKD staging criteria, advanced ML techniques, and XAI.

II. METHODS

This section outlines the methodology followed in the study, covering data sources, ethical considerations, data preparation, eGFR calculation, statistical analysis, predictive modeling, performance evaluation, and model explainability. Each subsection describes the steps to develop and validate the CKD prediction and classification models.

A. DATA SOURCE AND STUDY POPULATION

The study utilized data available from the UK Biobank, a comprehensive biological database and research resource that contains health data from over half a million participants. These individuals were recruited between 2006 and 2010, with an age range of 40 to 69 years at baseline [39], [40]. Data collection occurred across 22 assessment centers in England, Scotland, and Wales, capturing a broad representation of the UK population.

Various types of data were collected from participants through different methods, including verbal interviews, physical examinations, touchscreen questionnaires, biomedical assessments, and biological samples (blood, urine, and saliva). The dataset is diverse regarding ethnicity, with participants categorized into groups such as White British (most common), mixed, Asian or Asian British, Black or Black British, Chinese, and others. Further details can be found on their website [41].

Initially, the study involved a cohort of 20,927 participants. However, 5,116 individuals were excluded from further analysis due to missing demographic and clinical data, including SCr and SCysC. Therefore, the analyzed cohort consisted of 15,811 individuals from the UK Biobank, aged between 40 and 69 years, as illustrated in Figure 1.

The CKD stages were identified using the International Classification of Diseases version 10 (ICD-10) codes N18.1 through N18.5, which were extracted from hospital records (data field ID 41270, UK Biobank dataset). CKD is defined in five stages, ranging from stage 1 (normal kidney function with evidence of kidney damage), stage 2 (reflecting mild kidney dysfunction), stage 3 (indicating moderate

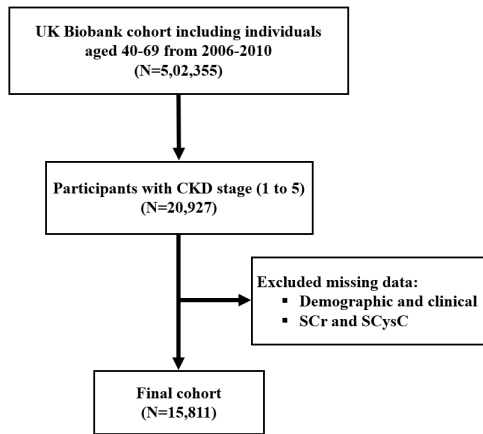


FIGURE 1. Flow diagram of the cohort study.

kidney dysfunction), stage 4 (severe kidney dysfunction), and stage 5 (kidney failure requiring dialysis or transplant).

The ICD system, endorsed by the World Health Organization (WHO), is a globally recognized diagnostic tool that ensures the systematic recording, analysis, interpretation, and comparison of mortality and morbidity data across regions and timeframes. The advanced stages of CKD (stages 3, 4, and 5) represent more severe kidney dysfunction and are associated with an increased risk of complications, including kidney failure. Participants were followed from the time of recruitment (between 2006 and 2010) until CKD diagnosis, death, or the conclusion of the follow-up period.

B. ETHICAL APPROVAL AND CONSENT TO PARTICIPATE

Ethical approval, emphasizing voluntary and well-documented participant involvement, was obtained from the National Information Governance Board for Health and Social Care and the North West Multicentre Research Ethical Committee as a Research Tissue Bank (RTB) approval (Ref: 11/NW/0382), which was extended in 2021 (Ref: 21/NW/0157). The study followed the Declaration of Helsinki principles, and all subjects gave informed consent via electronic signature upon recruitment.

C. DATA PREPARATION

The UK Biobank dataset includes comprehensive clinical and biochemical measures relevant to kidney function and overall health. A standardized biochemistry sampling, handling, and quality control protocol were previously validated [42]. Clinical data from the UK Biobank cohort were centrally processed, with blood and spot urine samples analyzed for key biomarkers such as SCr, SCysC, blood pressure (systolic and diastolic), glycated hemoglobin (HbA1c), albumin, and inflammatory markers like C-reactive protein (CRP).

A subset of 15,811 patients diagnosed with chronic kidney CKD was analyzed, incorporating demographic and clinical characteristics. The dataset includes laboratory measurements and vital signs essential for assessing kidney function.

Key demographic variables include age and body mass index (BMI), with a notable tendency toward overweight or obesity—risk factors for CKD progression. Biomarkers such as albumin (Alb) and CRP provide insights into nutritional status and systemic inflammation, while HbA1c reflects glycemic control, particularly in diabetes-related CKD. The eGFR was calculated using three CKD-EPI equations: SCr, SCysC, and a combined approach that integrates both markers. To enhance reliability, each patient had at least three eGFR measurements, ensuring a more accurate assessment of disease progression.

D. CALCULATIONS OF $EGFR_{SCr}$, $EGFR_{SCysC}$, $EGFR_{SCr-SCysC}$

Our study focuses on key variables, including SCr, SCysC, gender, and age, to analyze the complexities of equations for eGFR. This approach aims to elucidate the independent and combined roles of SCr and SCysC in diagnosing CKD. The eGFR was determined using the CKD-EPI method, which involved the calculation of eGFR utilizing SCr ($EGFR_{SCr}$), SCysC ($EGFR_{SCysC}$), or a combination of SCysC and SCr ($EGFR_{SCr-SCysC}$) equations, as follows:

$$eGFR_{SCr} = 142 \times \min(SCr/\kappa, 1)^\alpha \times \max(SCr/\kappa, 1)^{-1.200} \times 0.9938^{Age} \times (1.012 \text{ if 'female'}) \times (1.159 \text{ if 'African-descent'}) \quad (1)$$

$$eGFR_{SCysC} = 133 \times \min(SCysC/0.8, 1)^{-0.499} \times \max(SCysC/0.8, 1)^{-1.328} \times 0.996^{Age} \times (0.932 \text{ if 'female'}) \quad (2)$$

$$eGFR_{SCr-SCysC} = 135 \times \min(SCr/\kappa, 1)^\beta \times \max(SCr/\kappa, 1)^{-0.544} \times \min(SCysC/0.8, 1)^{-0.323} \times \max(SCysC/0.8, 1)^{-0.778} \times 0.9961^{Age} \times (0.963 \text{ if 'female'}) \quad (3)$$

where SCr is the serum Cr level; SCysC is the serum CysC level; κ is 0.9 for 'male' and 0.7 for 'female'; α is -0.302 for 'male' and -0.241 for 'female'; and β is -0.144 for 'male' and -0.219 for 'female' [43], [44], [45]. A factor of 1.0 was assigned for ethnicity because there were no subjects of 'African descent' in this study.

Each eGFR value was classified into one of five groups based on the KDIGO staging of CKD (stages 1 to 5), determined by specific eGFR thresholds. Samples with an eGFR exceeding $60 \text{ mL/min/1.73m}^2$ were categorized as early stages, whereas those with an eGFR below this threshold ($< 60 \text{ mL/min/1.73m}^2$) were classified as late-stage CKD [46]. Figure 2 illustrates the distribution of eGFR categories based on the CKD-EPI_{SCr}, CKD-EPI_{SCysC}, and CKD-EPI_{SCr-SCysC} equations. These categories are defined as follows: CKD stage 1 corresponds to $eGFR \geq 90 \text{ mL/min/1.73m}^2$, CKD stage 2 includes values between 60 and $89.99 \text{ mL/min/1.73m}^2$, CKD stage 3 ranges from 30 to $59.99 \text{ mL/min/1.73m}^2$, CKD stage 4 spans 15 to

29.99 mL/min/1.73m², and CKD stage 5 is defined by an eGFR below < 15 mL/min/1.73m².

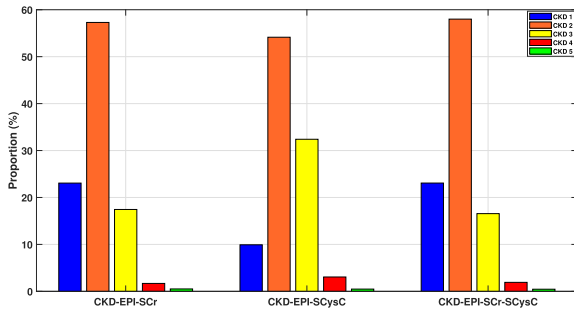


FIGURE 2. Distribution of patients across eGFR categories.

E. STATISTICAL ANALYSIS

Participants' demographic and clinical baseline characteristics were summarized as mean and standard deviation (SD) for continuous variables and percentages for categorical variables. Pairwise Pearson correlation analysis was conducted to evaluate relationships among individual features, including a comparative study of the three eGFR equations. Statistical significance was determined at a threshold of $p \leq 0.05$. All statistical analyses were conducted using Python 3.11.11. A Bland-Altman plot analysis was performed to assess agreement between the three CKD-EPI equations further, evaluating mean differences and potential biases in eGFR estimation [47]. This method provided insight into the interchangeability and reliability of the equations across different CKD stages.

F. PREDICTION MODEL DEVELOPMENT

The development of the prediction model follows a structured pipeline consisting of three main steps: (1) regression-based estimation of eGFR using different CKD-EPI equations, (2) optimization of model performance using the Grey Wolf Optimizer (GWO), and (3) classification of CKD stages using multiple ML classifiers. The detailed methodology is outlined below.

1) REGRESSION-BASED EGFR ESTIMATION

The first stage of model development focuses on predicting the eGFR using three widely accepted CKD-EPI equations. To evaluate their performance, two regression models were employed: Linear Regression (LR) [48] and Support Vector Regression (SVR) [49]. LR, a fundamental statistical approach, establishes a direct linear relationship between independent clinical features and eGFR, offering an interpretable framework for understanding basic trends in the data. However, SVR was implemented to capture more complex, non-linear relationships, leveraging kernel functions to optimize predictive performance. Unlike LR, SVR maintains a wide margin between predicted and actual values, effectively minimizing errors and improving estimation accuracy. These models were assessed to determine

which CKD-EPI equation provides the most reliable eGFR predictions before further optimization and classification.

2) OPTIMIZATION OF MODEL PERFORMANCE USING GWO

The GWO was employed as an optimization technique to enhance the predictive accuracy of both regression models. GWO is a nature-inspired metaheuristic algorithm that simulates grey wolves' leadership hierarchy and hunting strategies to achieve optimal solutions [50]. This study utilized GWO for hyperparameter tuning to enhance the predictive performance of LR and SVR models. By optimizing key parameters such as regularization terms (L1, L2), fit intercept, and solver choices in LR, as well as the learning rate, regularization terms, and kernel functions in SVR, GWO improved the predictive accuracy of both LR and SVR models. This optimization process enhanced model efficiency and ensured the selection of the most accurate CKD-EPI equation for subsequent CKD stage classification.

3) MULTI-CLASS CKD STAGE CLASSIFICATION

Once the most suitable CKD-EPI equation for eGFR estimation was identified, the next step involved classifying CKD stages based on the eGFR values, which were categorized into five stages following standard clinical thresholds. To achieve accurate classification, four ML models were implemented: Support Vector Machine (SVM) [51], Decision Tree (DT) [52], Random Forest (RF) [53], and Extreme Gradient Boosting (XGBoost) [54]. SVM, known for its effectiveness in high-dimensional datasets, utilized kernel functions to construct optimal hyperplanes for CKD stage separation. DT, a rule-based approach, provided an interpretable model by creating a tree-like structure based on decision rules, though it is susceptible to overfitting. RF, an ensemble learning technique, mitigated overfitting by aggregating multiple decision trees, enhancing overall classification performance. XGBoost, a robust gradient-boosting algorithm, further improved classification accuracy by sequentially combining weak models into a stronger predictive framework. Each classifier was trained using eGFR estimates from the selected CKD-EPI equation to determine the most effective model for CKD stage classification.

G. PERFORMANCE EVALUATION

The performance of the developed models was assessed based on various statistical and ML evaluation metrics. The evaluation process was divided into two main parts: (1) regression model evaluation for eGFR estimation and (2) classification model evaluation for CKD stage classification.

1) REGRESSION MODEL EVALUATION

To compare the predictive capability of different CKD-EPI equations, the performance of the LR and SVR models was evaluated using five standard regression metrics: root mean squared error (RMSE), mean absolute

percentage error (MAPE), ratio of standard deviation to RMSE (RSR), coefficient of determination (R^2), and Willmott's index of agreement (WIA). RMSE measures the standard deviation of prediction errors, while MAPE provides a percentage-based measure of prediction performance. RSR normalizes RMSE using the standard deviation of actual values, whereas R^2 quantifies the proportion of variance explained by the model. WIA assesses model performance by comparing squared errors against potential errors. The CKD-EPI equation that achieved the lowest RMSE and highest R^2 score was considered the most reliable for eGFR estimation. Additionally, the impact of the GWO on model performance was analyzed by comparing results before and after optimization to assess its effectiveness in improving predictive performance.

2) CLASSIFICATION MODEL EVALUATION

Once the most suitable CKD-EPI equation was identified, the next step involved evaluating the performance of ML classifiers SVM, DT, RF, and XGBoost for CKD stage classification. The assessment was conducted using multiple evaluation metrics to ensure reliability and robustness. Accuracy measured the overall proportion of correctly classified CKD stages, while precision evaluated the reliability of positive predictions by determining the ratio of correctly identified cases to total predicted cases in each stage. Recall (sensitivity) assessed the model's ability to correctly detect CKD cases by comparing correctly classified instances against actual cases. To provide a balanced measure, the F1-score, a harmonic mean of precision and recall, was used to account for false positives and false negatives.

Additionally, the area under the Receiver Operating Characteristic curve (AUC-ROC) was employed to evaluate the classifier's ability to distinguish between different CKD stages, with higher values indicating superior discrimination. To enhance the robustness of the evaluation, cross-validation was performed, ensuring the model's generalizability across various subsets of data. The best-performing model was selected for CKD staging based on the overall classification performance and clinical relevance.

H. MODEL EXPLAINABILITY

Shapley Additive Explanations (SHAP) enhance the interpretability of model predictions. Derived from cooperative game theory, SHAP calculates the contribution of each feature to a model's prediction by considering all possible permutations of feature values. By quantifying the impact of individual features, SHAP provides valuable insights into the relative importance of clinical variables, ensuring that ML models remain transparent, clinically meaningful, and actionable [38].

This study applies SHAP for global and local interpretations of the CKD stage prediction model. For global interpretation, SHAP feature importance plots assess the overall impact of each feature across all predictions. SHAP waterfall plots explain individual predictions for local

interpretation, illustrating how the model classifies instances of varying CKD stages.

III. RESULTS

This study included 15,811 participants diagnosed with CKD, with a mean age of 62.22 ± 6.07 years. The average age was slightly higher in males (62.43 ± 6.01 years) compared to females (61.98 ± 6.13 years). The gender distribution of the cohort was 8,216 for males (51.96%) and 7,595 for females (48.04%). A detailed summary of the demographic characteristics and laboratory data is presented in Table 1. To explore potential associations among key

TABLE 1. Characteristics of 15,811 patients diagnosed with CKD, including demographic and clinical data.

Characteristics	Mean $\pm$ SD
Age (years)	62.22 ± 6.07
BMI (Kg/m^2)	29.69 ± 5.38
Alb (g/L)	44.38 ± 2.84
HbA1c (mmol/mol)	40.65 ± 11.43
SBP (mmHg)	144.72 ± 21.99
DBP (mmHg)	81.15 ± 11.79
C-reactive protein (mg/L)	3.89 ± 5.69
SCr (mg/dL)	1.08 ± 0.57
SCysC (mg/L)	1.19 ± 0.44
eGFR CKD-EPI _{SCr} ($\text{mL}/\text{min}/1.73\text{m}^2$)	74.93 ± 18.87
eGFR CKD-EPI _{SCysC} ($\text{mL}/\text{min}/1.73\text{m}^2$)	65.89 ± 18.93
eGFR CKD-EPI _{SCr-SCysC} ($\text{mL}/\text{min}/1.73\text{m}^2$)	75.71 ± 19.33

physiological markers, a comprehensive correlation analysis was conducted, examining variables such as BMI, Alb, HbA1c, SBP, DBP, CRP, SCr, and SCysC. The results of this analysis are summarized in Figure 3, providing insights into biomarker interactions relevant to CKD progression.

A correlation matrix heatmap was generated to visualize Pearson's correlation coefficients between clinical variables, including SCr, SCysC, and other biomarkers. The most notable finding was the strong positive correlation between SCr and SCysC ($r = 0.849$), indicating that these two markers tend to increase or decrease together, reinforcing their role as key indicators of kidney function. Other observed relationships were generally weaker, with only a few moderate correlations. For example, SBP and DBP showed a moderate positive correlation ($r = 0.449$), which aligns with their physiological connection.

In contrast, several weak correlations were identified, such as those between SCr and BMI ($r = 0.017$), SCr and CRP ($r = 0.059$), and SCysC and BMI ($r = 0.126$), suggesting minimal direct relationships. Additionally, notable negative correlations were observed, including Alb and CRP ($r = -0.209$) and Alb and SCysC ($r = -0.185$), implying that higher albumin levels were associated with slightly lower CRP and SCysC levels. These findings indicate that CKD progression is influenced by multiple interrelated factors, emphasizing the need for a comprehensive, multi-biomarker approach in disease monitoring and management.

To further explore these associations, eGFR values reflecting different CKD stages were calculated using the CKD-EPI

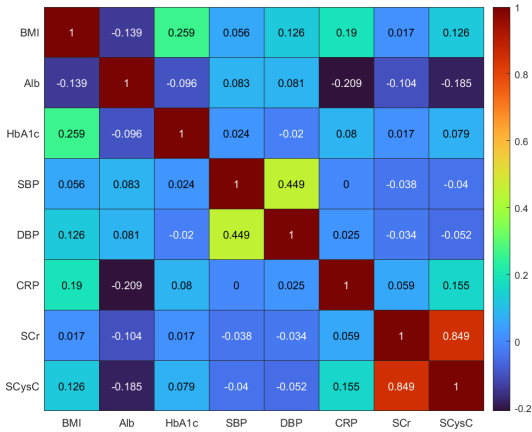


FIGURE 3. Correlation matrix of (SCr and SCysC) and clinical variables. Correlation strength (r) is categorized as weak ($|r| < 0.3$), moderate ($0.3 \leq |r| < 0.7$), and strong ($|r| \geq 0.7$), with statistically significant correlations defined as $p < 0.05$. The color scale represents the strength and direction of the correlation, with red indicating strong positive correlations, blue representing weak or negative correlations, and intermediate colors showing moderate associations.

equation. These values were derived from SCr, SCysC, and their combination, as presented in Table 2.

TABLE 2. eGFR values using CKD-EPI equation in different stages of CKD based on SCr, SCysC, and combined SCr and SCysC.

CKD Stage	eGFR _{SCr}	eGFR _{SCysC}	eGFR _{SCr-SCysC}
CKD 1	98.14 ± 5.42	98.56 ± 6.58	98.83 ± 6.98
CKD 2	75.21 ± 8.05	73.01 ± 8.09	76.37 ± 8.12
CKD 3	49.65 ± 7.79	48.81 ± 8.03	48.92 ± 7.88
CKD 4	24.45 ± 4.01	23.89 ± 3.95	24.18 ± 3.91
CKD 5	8.67 ± 3.17	9.50 ± 2.87	8.88 ± 2.75
p-value	< 0.001	< 0.001	< 0.001

This table compares eGFR values calculated using these three different methods across various stages of CKD. As CKD progresses from stage 1 to stage 5, the eGFR values consistently decline across all three methods, indicating a deterioration in kidney function. For CKD stage 1, the mean eGFR values are similar across methods (approximately 98 mL/min/1.73m²). However, these values drop significantly as the disease advances, with CKD 5 showing mean eGFR values around 8-9 mL/min/1.73m².

The extremely small p -values (all less than 0.05) confirm that the differences in eGFR between CKD stages are statistically significant. This finding highlights the effectiveness of the CKD-EPI equation, whether based on SCr, SCysC, or their combination, in distinguishing CKD stages. The consistent decline in eGFR values across all methods further reinforces their reliability for assessing kidney function and accurately staging CKD. Significant differences are observed at each stage, emphasizing the robustness of these approaches.

Figure 4 presents boxplots illustrating eGFR across CKD stages using different equations, visually representing its distribution and variability within each stage. Additionally,

statistical analyses confirm that eGFR differences between all stage pairs remain significant across the evaluated equations, with p -values consistently below 0.001, further validating the effectiveness of these methods in distinguishing CKD stages.

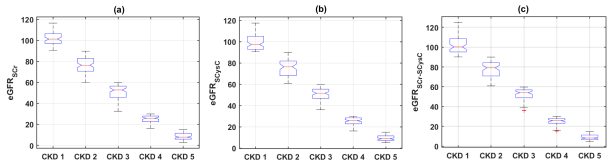


FIGURE 4. (a) Boxplot for eGFR_{SCr} for all CKD stages, (b) Boxplot for eGFR_{SCysC} for all CKD stages, (c) Boxplot for eGFR_{SCr-SCysC} for all CKD stages.

In evaluating the correlation of eGFR with biomarkers, a stronger negative correlation was observed between eGFR and SCysC compared to SCr, as detailed in Table 3. Notably,

TABLE 3. Correlation between SCr and/or SCysC with three frequently used eGFR equations.

Parameters	Equations Applied	r	p
SCr vs. eGFR _{SCr}	CKD-EPI _{SCr} (2021)	-0.6772	< 0.01
SCysC vs. eGFR _{SCysC}	CKD-EPI _{SCysC} (2012)	-0.8015	< 0.01
SCysC vs. eGFR _{SCr-SCysC}	CKD-EPI _{SCr-SCysC} (2021)	-0.8253	< 0.01

SCysC exhibited the highest correlation with eGFR ($r = -0.8253$) in the combined equation CKD-EPI_{SCr-SCysC}, surpassing the individual performance of both SCysC alone and SCr-based eGFR in the detection of renal impairment. This suggests that SCysC, particularly in combination with SCr, provides a more accurate reflection of kidney function.

Additionally, the relationship between SCr and SCysC is graphically depicted in Figure 5(a), which illustrates their correlation (Pearson correlation coefficient = 0.849). This scatter plot shows how tightly data points cluster around an upward-sloping diagonal line, indicating a strong positive correlation between the two biomarkers. A weaker correlation would result in more scattered data points, potentially influenced by external factors such as muscle mass or age, which can affect marker metabolism and clearance. Clinicians can use this graphical analysis to understand kidney health and disease progression better.

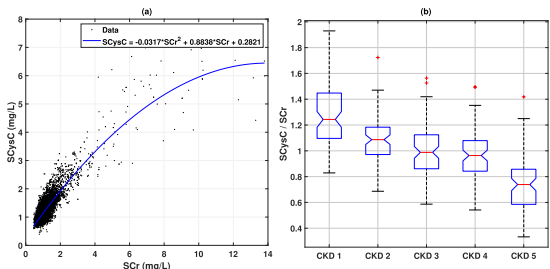


FIGURE 5. (a) Quadratic regression between SCr and SCysC, (b) Boxplot for SCysC to SCr ratio according to CKD stages.

Furthermore, Figure 5(b) presents a box plot graph illustrating the SCysC to SCr ratio across different CKD

stages. This plot highlights a significant decline in the SCysC/SCr ratio in advanced CKD stages, offering clinicians valuable insights into assessing kidney health and monitoring CKD progression. Table 4 provides the statistical analysis of the SCysC/SCr ratio, revealing significant differences between most stages ($p < 0.05$).

TABLE 4. Results of statistical analysis for calculated ratio across each pair of the CKD stages.

Calculated Ratio	CKD 1	CKD 2	CKD 3	CKD 4	CKD 5	p-value
SCysC/SCr	✓	✓	-	-	-	< 0.001*
	✓	-	✓	-	-	< 0.001*
	✓	-	-	✓	-	< 0.001*
	✓	-	-	-	✓	< 0.001*
	-	✓	✓	-	-	0.0076*
	-	✓	-	✓	-	< 0.001*
	-	-	✓	-	✓	< 0.001*
	-	-	✓	✓	-	0.2486
	-	-	-	✓	✓	< 0.001*
	-	-	-	-	✓	< 0.001*

* $p < 0.05$ was considered statistically significant.

The ratio notably distinguishes CKD stage 1 from later stages ($p < 0.001$), though no significant difference is observed between stages 3 and 4, suggesting that kidney function is similar at these mid-stages. Figure 6 illustrates the distribution of eGFR within a sample population, calculated using different CKD-EPI equations based on the biomarkers: (a) SCr, (b) SCysC, and (c) a combination of both SCr and SCysC.

This comparison highlights how eGFR values vary depending on the biomarker used, providing insights into their sensitivity in detecting kidney dysfunction. The median eGFR based on SCr is 75.93 mL/min/1.73m², while SCysC alone yields a lower median of 66.45 mL/min/1.73m², suggesting potential differences in biomarker sensitivity. However, when both SCr and SCysC are combined, the median eGFR increases to 77.80 mL/min/1.73m², indicating that integrating both biomarkers may enhance eGFR estimation accuracy.

To supplement the correlation analysis, Bland-Altman analysis was performed to assess the agreement among the CKD-EPI_{SCr}, CKD-EPI_{SCr-SCysC}, and CKD-EPI_{SCysC} equations for eGFR estimation. While correlation measures the strength of the association between these methods, it does not evaluate whether the equations yield comparable and interchangeable eGFR values. Bland-Altman analysis provides a more rigorous assessment by quantifying systematic biases and the extent of variability between estimation methods.

As shown in Figure 7, the x -axis represents the mean eGFR of the two compared equations, while the y -axis displays their difference. The red line denotes the mean difference (bias), and the blue dashed lines indicate the limits of agreement (± 1.96 SD). Figure 7(a) compares CKD-EPI_{SCr} and CKD-EPI_{SCr-SCysC}, showing a mean difference of 0.8388 (SD = 9.9751, and a confidence interval (CI) = [0.6833, 0.9943]), suggesting minimal bias and strong agreement. Figure 7(b) contrasts CKD-EPI_{SCr} with CKD-EPI_{SCysC}, revealing a mean difference of -8.9471

(SD = 13.6434, CI = [-9.1598, -8.7345]), indicating that CKD-EPI_{SCysC} systematically underestimates eGFR, with worsening agreement at higher eGFR values. Figure 7(c) compares CKD-EPI_{SCr-SCysC} and CKD-EPI_{SCysC}, showing a mean difference of -9.7859 (SD = 6.1276, CI = [-9.8814, -9.6904]), suggesting that CKD-EPI_{SCysC} consistently estimates lower eGFR values.

Overall, CKD-EPI_{SCr-SCysC} aligns closely with CKD-EPI_{SCr}, making them more interchangeable, while CKD-EPI_{SCysC} tends to underestimate eGFR, which may impact CKD classification, particularly in individuals with higher eGFR values. These findings underscore the need to carefully select the eGFR estimation method to optimize clinical decision-making and ensure accurate kidney function assessment.

This study estimated the GFR using three different CKD-EPI equations, forming the foundation for subsequent modeling and analysis. A regression analysis was conducted to determine the most accurate equation for eGFR estimation, incorporating features selected through Pearson correlation analysis and Variance Inflation Factor (VIF) assessment [55]. The dataset underwent thorough preprocessing to ensure consistency and optimize model performance. Since no missing values were present, categorical and numerical variables were handled appropriately. The categorical variable "Sex" (Male/Female) was encoded as a binary variable (Male = 1, Female = 0) to align with the requirements of prediction models, while numerical features were standardized using z -score normalization to enhance model stability and convergence.

Feature selection was performed using Pearson correlation and VIF analysis, ensuring the retention of only the most relevant features while mitigating potential multicollinearity. SCr and SCysC exhibited strong negative correlations with eGFR, making them key predictors. The VIF values for all features remained well below the critical threshold of 10, confirming the absence of multicollinearity issues, as shown in Table 5. Consequently, all features, including Age, BMI, SCr, SCysC, HbA1c, CRP, Alb, SBP, and DBP, were retained for further analysis and model training.

TABLE 5. VIF values for features.

Feature	VIF Value
Age	1.08
Sex	1.06
BMI	1.17
Creatinine	3.96
Cystatin C	4.05
HbA1c	1.09
C-reactive protein	1.10
Albumin	1.10
SBP	1.34
DBP	1.32

Following feature selection, the dataset was split into training (70%) and testing (30%) sets to facilitate model evaluation. LR and SVR models were trained using the

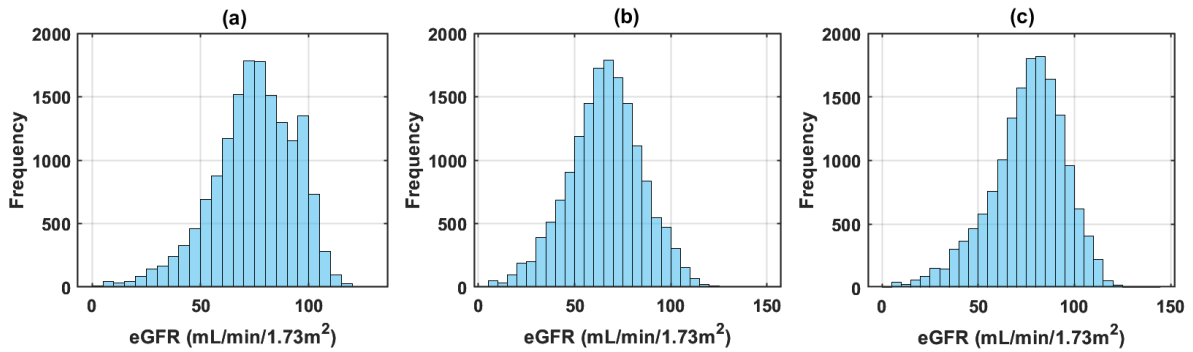


FIGURE 6. Distribution of eGFR calculated based on different CKD-EPI equations for biomarkers: (a) SCr, (b) SCysC, and (c) Combined SCr and SCysC.

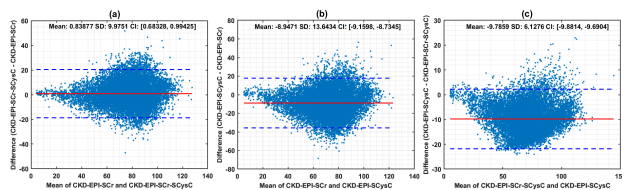


FIGURE 7. Bland-Altman plots for (a) CKD-EPI_{SCr}, (b) CKD-EPI_{SCysC}, and (c) CKD-EPI_{SCr-SCysC} equations.

refined feature set, with the three eGFR estimates derived from the CKD-EPI equations serving as dependent variables. The GWO was further integrated into LR and SVR models to enhance predictive accuracy, resulting in LR-GWO (Linear Regression with GWO) and SVR-GWO (Support Vector Regression with GWO). The GWO algorithm, inspired by the leadership and hunting strategies of grey wolves, optimizes hyperparameters to improve model convergence and predictive accuracy.

The implementation of GWO followed a structured approach: initially, a population of search agents (wolves) was randomly initialized within predefined parameter bounds. The fitness of each agent was evaluated based on the model's prediction error. The best three solutions—designated as α (alpha), β (beta), and δ (delta) wolves—guided the rest of the population toward the optimal solution using adaptive coefficient vectors that simulate encircling, attacking, and searching behaviors.

In LR-GWO, GWO was applied to optimize regression coefficients and regularization parameters by minimizing the loss function. In SVR-GWO, it fine-tuned hyperparameters such as the regularization parameter (C), kernel function parameters (γ), and error sensitivity parameter (ϵ). The optimized SVR-GWO model utilized $C = 5.05$, $\epsilon = 0.5$, $\gamma = 0.06$, and an 'rbf' kernel. The optimization was performed using 40 search agents, 3 iterations, and 3 leadership groups (α , β , δ) for parameter tuning. This approach ensured effective exploration and exploitation of the search space, improving the predictive models' generalization performance.

The performance comparison of prediction models, including LR, SVR, GWO-Optimized LR, and GWO-Optimized SVR, across different CKD-EPI equations, is presented in Table 6. The results indicate that SVR-based models con-

TABLE 6. Performance Comparison of Prediction Models (LR, SVR, GWO-Optimized LR, and GWO-Optimized SVR) Based on Various Metrics for Each CKD-EPI Equation.

Prediction Models	eGFR Equations	RMSE	MAPE (%)	RSR	R ²	WIA
LR	CKD-EPI _{SCr}	12.44	17.11	0.67	0.55	0.81
	CKD-EPI _{SCysC}	9.55	14.09	0.51	0.74	0.92
	CKD-EPI _{SCr-SCysC}	10.22	13.92	0.53	0.71	0.91
SVR	CKD-EPI _{SCr}	8.51	10.19	0.46	0.79	0.94
	CKD-EPI _{SCysC}	3.43	3.95	0.18	0.97	0.99
	CKD-EPI _{SCr-SCysC}	3.03	2.97	0.16	0.97	0.99
LR-GWO	CKD-EPI _{SCr}	12.38	17.56	1.21	0.55	0.79
	CKD-EPI _{SCysC}	10.24	15.68	0.71	0.70	0.89
	CKD-EPI _{SCr-SCysC}	9.97	13.30	0.62	0.72	0.91
SVR-GWO	CKD-EPI _{SCr}	4.46	4.54	0.26	0.94	0.98
	CKD-EPI _{SCysC}	2.50	2.70	0.14	0.98	0.99
	CKD-EPI _{SCr-SCysC}	1.90	1.86	0.10	0.99	0.99

sistently outperformed LR-based models, with SVR-GWO achieving the highest predictive performance. Among the three CKD-EPI equations, the SCr-SCysC equation provided the best performance when paired with SVR-GWO, achieving RMSE = 1.90, MAPE = 1.86%, R² = 0.99, and WIA = 0.99.

In contrast, traditional LR models exhibited the highest errors, while LR-GWO showed slight improvements but remained inferior to SVR-based models. The integration of GWO in SVR optimization significantly enhanced prediction performance, reaffirming the effectiveness of automated hyperparameter tuning in improving model capability. Overall, SVR-GWO emerged as the most effective model for CKD-EPI-based eGFR estimation, mainly when using SCr-SCysC inputs. These findings underscore the advantages of incorporating metaheuristic optimization techniques like GWO to enhance predictive performance in medical modeling.

In this study, we implemented and compared four machine learning models, SVM, DT, RF, and XGBoost, for multi-class classification of CKD stages, ranging from 1 to 5. To ensure consistent feature scaling, all numerical features were standardized using StandardScaler. The dataset was

split into training and testing sets in a 70:30 ratio, with stratification applied to preserve class distribution. To address the inherent class imbalance in CKD stage classification, the **Synthetic Minority Over-sampling Technique (SMOTE)** was employed within a pipeline to generate synthetic samples for minority classes [56]. Importantly, SMOTE was applied exclusively to the training folds during cross-validation, thereby preventing data leakage. To further enhance model performance and ensure robust evaluation, we implemented **Stratified 5-Fold Cross-Validation (SKCV)** [57]. This approach maintains class proportions across all folds, reducing the likelihood of subsets with insufficient minority class instances. By mitigating variance, SKCV provides a more reliable estimate of model performance, particularly in imbalanced datasets. **Hyperparameter tuning** was performed using the grid search method to optimize model performance by identifying the best hyperparameter configurations. The specific configurations for each model are detailed in Table 7.

The effectiveness of the models was evaluated using key classification metrics, including accuracy, precision, recall, and F1-score, computed with `scikit-learn` and visualized using `matplotlib` and `Seaborn` in Python 3.11.11. Model performance was assessed on training

TABLE 7. Hyperparameter search space and selected values for ML models.

ML Model	Hyperparameter	Search space	Selected value
SVM	kernel	'linear', 'rbf', 'poly'	poly
	C	0.1, 1, 10, 100	1
	gamma	0.01, 0.1, 1	1
DT	Criterion	'gini', 'entropy'	entropy
	max_depth	10, 20, 30	20
	min_samples_split	2, 5, 10	2
	min_samples_leaf	1, 2, 4	4
RF	Criterion	'gini', 'entropy'	entropy
	n_estimators	50, 100, 200	100
	max_depth	10, 20, 30	20
	min_samples_split	2, 5, 10	2
	min_samples_leaf	1, 2, 4	2
XGBoost	n_estimators	50, 100, 200	200
	max_depth	10, 20, 30	20
	learning_rate	0.01, 0.1, 0.2	0.2
	subsample	0.8, 0.9, 1.0	0.8
	colsample_bytree	0.8, 0.9, 1.0	0.9
	gamma	0, 0.1, 0.2	0.2

and testing datasets, with confusion matrices generated to visualize class-wise prediction accuracy and multi-class ROC curves with Area Under the Curve (AUC) values computed using a one-vs-rest approach to provide a detailed performance analysis. A summary of the model performances is presented in Table 8. Among the models tested, XGBoost

TABLE 8. Performance evaluation of ML models.

Model	Phase	Accuracy (%)	Precision (%)	Sensitivity (%)	F1-Score (%)
SVM	Training	96.54 ± 0.18	96.63 ± 0.24	96.54 ± 0.15	96.55 ± 0.23
	Testing	93.66 ± 0.47	93.76 ± 0.28	93.70 ± 0.26	93.68 ± 0.39
DT	Training	96.89 ± 0.37	96.90 ± 0.01	96.87 ± 0.21	96.90 ± 0.09
	Testing	95.15 ± 0.29	95.17 ± 0.38	95.15 ± 0.14	95.14 ± 0.25
RF	Training	97.37 ± 0.08	97.40 ± 0.03	97.36 ± 0.31	97.42 ± 0.29
	Testing	96.19 ± 0.07	96.17 ± 0.08	96.24 ± 0.06	96.12 ± 0.02
XGBoost	Training	98.22 ± 0.02	98.20 ± 0.07	98.19 ± 0.15	98.17 ± 0.30
	Testing	97.76 ± 0.43	97.68 ± 0.07	97.64 ± 0.18	97.45 ± 0.08

received the highest F1-score of 97.45%, demonstrating its ability to properly balance precision (97.68%) and recall (97.64%) across all classes. After that, RF came in second with an F1 score of 96.12%, which was backed by precision (96.17%) and recall (96.24%). On the other hand, DT and SVM earned F1 scores of 95.14% and 93.68%, respectively. The dataset exhibits potential class imbalance, necessitating the use of the F1-score for performance evaluation. This metric effectively incorporates false positives and false negatives, rendering it a more dependable measure of model effectiveness than accuracy alone.

The results demonstrate that **XGBoost consistently surpassed other models regarding overall classification performance**, establishing it as the most effective option among the algorithms evaluated. The F1-score was utilised as a critical metric for assessing model performance, given the potential class imbalance present in the dataset. This metric effectively accounts for both false positives and false negatives. This provides a more reliable metric than accuracy alone for evaluating the true effectiveness of each model. The results demonstrate that **XGBoost consistently surpassed other models regarding overall classification performance**, establishing it as the most effective option among the evaluated algorithms.

The confusion matrix and ROC curve, shown in Fig. 8, further validated XGBoost's strong classification performance, with high AUC values across all CKD stages. The model demonstrated minimal misclassifications, with a particularly strong performance in classifying CKD stage 2, which had the highest correctly classified instances. The ROC curve confirmed the model's robustness, with AUC values close to 1.0, indicating an excellent ability to distinguish between different disease stages. The superior performance of XGBoost can be attributed to its boosting strategy, where trees are built sequentially, correcting errors from previous models to enhance predictive efficiency. This makes XGBoost a highly scalable and efficient end-to-end tree-boosting system, particularly well-suited for handling complex medical datasets. Given the increasing demand for individualized and precise treatments, XGBoost's rapid execution speed and strong generalization capability position it as a leading ML alternative for disease classification.

The feature importance ranking derived from SHAP values for XGBoost is presented in Fig. 9, highlighting the key predictors influencing model decisions. The horizontal stacked bar chart effectively illustrates the relative importance of various features in predicting CKD stages by displaying features on the y-axis and their corresponding mean absolute SHAP values on the x-axis. The length of each bar reflects the overall influence of a feature on model predictions, while the colored segments within the bars indicate the contribution of each feature to specific CKD stages (1 to 5). Among the features analyzed, CysC and Cr exhibit the highest importance, significantly impacting model predictions, while Age, Sex, and BMI show moderate influence. In contrast, DBP, SBP, HbA1c, Alb, and CRP contribute minimally.

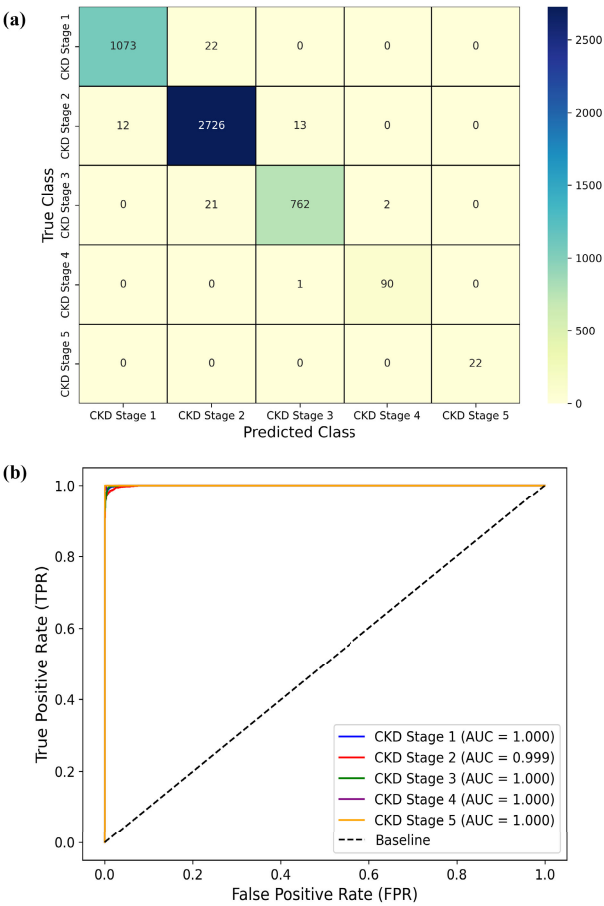


FIGURE 8. (a) Confusion matrix of the XGBoost model's predicted results in the test data, (b) Receiver Operating Characteristic (ROC) curve for the XGBoost model, displaying AUC values for each CKD stage.

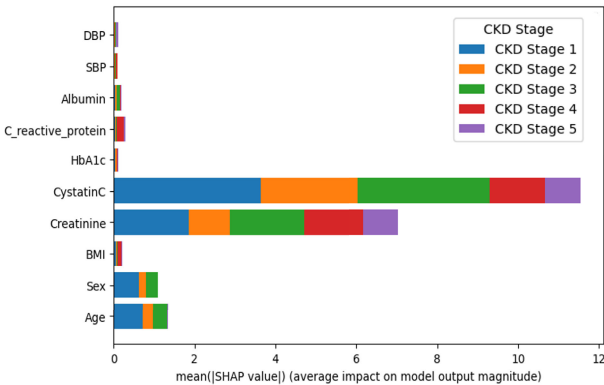


FIGURE 9. Mean absolute SHAP value per feature across CKD stages.

The color-coded stacked bars represent how each feature's impact varies across different stages of CKD, offering valuable insights into disease progression. For instance, a large segment corresponding to CKD stage 5 within the Creatinine bar suggests that this feature significantly predicts advanced kidney diseases. This visualization is crucial for identifying stage-specific predictors, understanding the

global importance of features, and ensuring interpretability in clinical applications.

To enhance the interpretability of our model's predictions, we utilized SHAP force analysis, which provides detailed, instance-level explanations by illustrating how each feature influences an individual prediction. To demonstrate the model's decision-making process, we selected five representative samples from the dataset, each diagnosed with CKD at different stages. Figure 10 presents SHAP force plots for these five individualized predictions, visually representing how various features contribute to the model's output. The

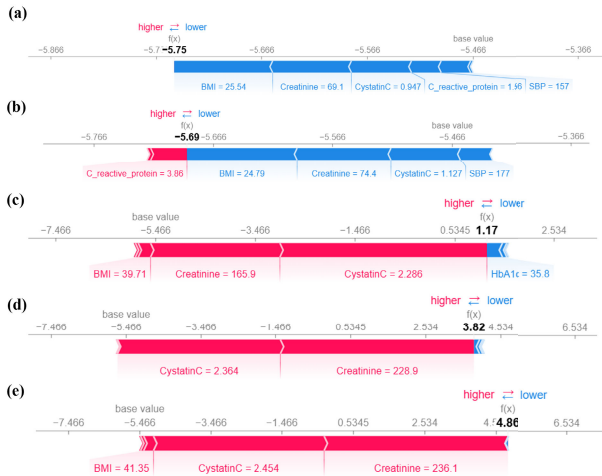


FIGURE 10. SHAP force plots illustrating feature contributions to model predictions for patients diagnosed with CKD at different stages (a) Patient ID: 10527 (stage 1), (b) Patient ID: 14642 (stage 2), (c) Patient ID: 6511 (stage 3), (d) Patient ID: 15590 (stage 4), (e) Patient ID: 14887 (stage 5).

horizontal axis represents the model's prediction value $f(x)$, with the base value (expected model output) indicated. Features that increase the prediction (positive contributions) are highlighted in red, while those that lower it (negative contributions) appear in blue. The magnitude of each bar reflects the strength of a feature's impact on the final prediction.

Each subplot in Figure 10(a-e) corresponds to a specific sample ID and CKD stage, illustrating how individual features shape the model's decision. In the earlier CKD stages (Figure 10a-b), features such as BMI, Cr, and CysC predominantly contribute negatively, reducing the prediction. However, as CKD progresses (Figure 10c-e), key biomarkers like CRP, Cr, and CysC exert a stronger positive influence, driving the prediction higher. This visualization offers an intuitive understanding of how different risk factors impact the model's predictions across various stages of CKD, enhancing transparency and interpretability in its decision-making process.

To further analyze how individual features impact CKD stage predictions, we employed SHAP waterfall plots, as illustrated in Figure 11, which play a crucial role in multiclass classification by breaking down feature contributions to individual predictions. Unlike binary classification,

where interpretation is relatively straightforward, multiclass classification involves multiple possible labels, making the decision process more complex. The SHAP waterfall plot provides a structured breakdown of how each feature influences the prediction, offering critical insights into medical diagnostics where interpretability is paramount.

Each subplot in Figure 11(a–e) represents a different patient, with $f(x)$ denoting the model's prediction and $E[f(x)] = -5.466$ indicating the expected model output across the dataset. Patient ID 10527 and Patient ID 14642 (Figures 11a and 11b) exhibit negative predictions ($f(x) < 0$), primarily driven by BMI, Cr, and CysC, which contribute to lowering the prediction score. In contrast, CRP has a minor positive effect, exerting only a slight upward influence on the prediction. These patients exhibit relatively lower levels of Cr and CysC, resulting in reduced risk predictions. Patient ID 6511 (Figure 11c) has a slightly positive prediction ($f(x) = 1.169$), driven by high levels of creatinine and CysC, which significantly increase the model's output. Patient ID 15590 (Figure 11d) shows a stronger positive prediction ($f(x) = 3.816$), largely driven by Cr and CysC, with minor contributions from BMI and other factors. Patient ID 14887 (Figure 11e) has the highest prediction ($f(x) = 4.859$), where Cr and CysC are the most significant positive contributors, with additional minor contributions from BMI, CRP, and Alb, while SBP and DBP have minimal effects.

These observations indicate that Cr and CysC play a dominant role in patients with high predictions, suggesting that the model associates higher levels of these biomarkers with increased CKD severity, whereas patients with negative predictions are influenced more by BMI, SBP, and DBP, which contribute to lowering the score. The model's reliance on Cr and CysC suggests that it effectively identifies these biomarkers as strong indicators of CKD progression, making SHAP-based visualizations crucial for enhancing model transparency and aiding clinical interpretation of CKD stage predictions.

IV. DISCUSSION

This study explored the relationships between clinical biomarkers and CKD progression in a large cohort of 15,811 patients, highlighting the role of key physiological markers such as SCr, SCysC, Alb, and blood pressure in evaluating kidney function. The findings revealed several significant trends that align with existing literature on CKD progression while providing novel insights into the utility of combining biomarkers and leveraging advanced computational techniques for improved prediction.

The study cohort exhibited a typical distribution for CKD, with a mean age of 62.22 years and a slightly higher proportion of males, reflecting the known trend of CKD being more prevalent in older adults and males. The elevated BMI (29.69 ± 5.38 Kg/m²) in the cohort underscores obesity as a significant risk factor for CKD. Furthermore, the elevated systolic (144.72 ± 21.99 mmHg) and diastolic (81.15 ± 11.79 mmHg) blood pressures observed align with

the well-established association between hypertension and CKD, emphasizing the importance of effective blood pressure management in CKD patients.

Our study found the strongest positive relationship between SCysC and eGFR in the CKD-EPI_{SCr-SCysC} combined equation ($r = -0.8253$). This was followed by SCysC alone ($r = -0.8015$) and SCr-based equations ($r = -0.6772$), as shown in Table 3. The correlations between SCr and SCysC ($r = 0.849$) confirm previous findings, indicating that these biomarkers often behave similarly, reflecting kidney function in tandem. In contrast, weaker correlations between BMI and kidney biomarkers suggest that, while BMI is an important factor, its direct influence on kidney function may not be as prominent as other indicators, such as SCr and SCysC.

Additionally, the negative correlations between albumin and inflammatory markers like CRP and SCysC highlight the role of inflammation in CKD progression, further supporting the importance of monitoring albumin levels as an indicator of kidney health. The Bland-Altman plots presented in Figure 7 offer a nuanced assessment of agreement among eGFR equations, revealing potential biases and discrepancies between them. These findings emphasize the importance of selecting appropriate equations based on the clinical context.

Regarding kidney function assessment, the significant decline in eGFR observed across all stages of CKD emphasizes the critical role of eGFR in tracking disease progression. Whether utilizing SCr, SCysC, or a combination of both, the CKD-EPI equations proved reliable for discriminating between CKD stages, confirming their relevance in clinical practice for diagnosing and staging CKD. Our findings support the utility of these equations, which continue to be widely used for patient management.

While traditional biomarker-based analyses remain important, this study extends the body of knowledge by incorporating ML models to predict eGFR and classify CKD stages. Applying ML, especially SVR, significantly improved prediction accuracy over traditional LR models. Notably, integrating GWO with SVR (SVR-GWO) enhanced performance, with an impressive RMSE of 1.90, MAPE of 1.86%, and R^2 of 0.99. This highlights the potential of GWO as an optimization technique that can refine predictive models and improve their performance. The improved accuracy of the SVR-GWO model emphasizes the value of combining traditional biomarkers with cutting-edge optimization techniques to make better predictions for CKD diagnosis and management.

Our study also explored ML models for CKD stage classification, including SVM, DT, RF, and XGBoost. The XGBoost model outperformed all others, achieving the highest F1 score of 97.45%. This finding reinforces the increasing evidence that ML models, especially ensemble methods like XGBoost, are highly effective in managing the complex, multidimensional data commonly associated with CKD diagnosis and prognosis. SHAP-based feature importance analysis adds a novel dimension to this research,

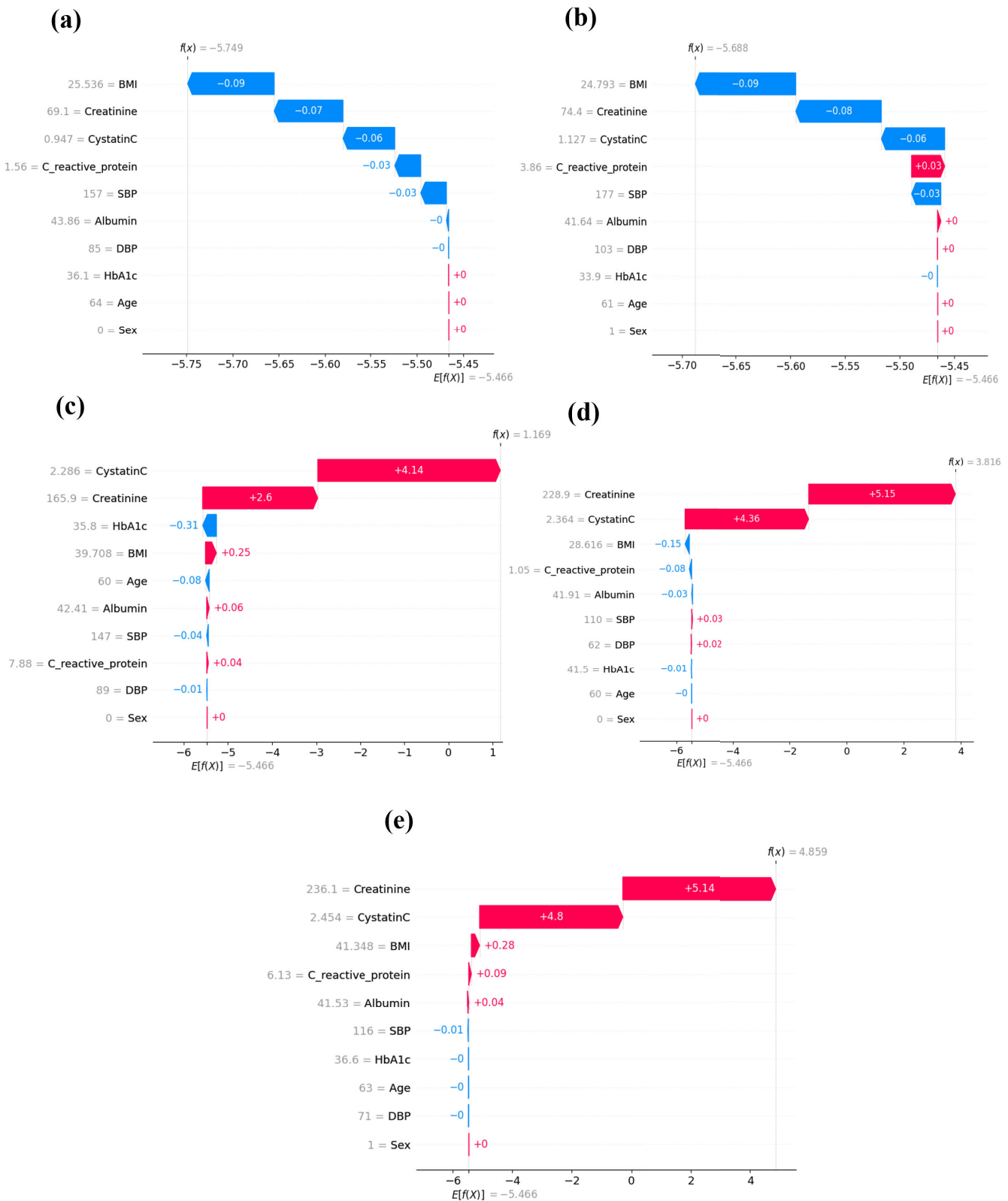


FIGURE 11. SHAP waterfall plots illustrating the contribution of each feature to model predictions for patients at CKD stages 1 to 5: (a) Patient ID 10527 (stage 1), (b) Patient ID 14642 (stage 2), (c) Patient ID 6511 (stage 3), (d) Patient ID 15590 (stage 4), (e) Patient ID 14887 (stage 5).

allowing for a more transparent and interpretable understanding of how specific biomarkers influence CKD predictions. By identifying the most influential features, we can offer accurate and clinically actionable insights, aligning with the growing emphasis on explainability in healthcare-related ML models.

Importantly, the findings also highlight the limitations of traditional eGFR-based equations, such as the Modification of Diet in Renal Disease (MDRD), Cockcroft-Gault (CG), and CKD-EPI. These equations overestimate GFR in patients with reduced kidney function ($\text{eGFR} \leq 60 \text{ mL/min/1.73m}^2$), potentially leading to misclassification of CKD stages and delayed diagnosis [17], [19]. Our results are consistent with recent studies that emphasize the need for more accurate, individualized approaches to eGFR estimation, especially for high-risk populations such as those with diabetes or hypertension, where early detection is crucial for effective intervention.

Combining biomarkers (SCr and SCysC) with ML optimization is a promising advance in predicting and staging CKD. Our findings suggest that ML models, when optimized with appropriate eGFR equations, offer superior predictive accuracy and robustness compared to conventional CKD staging methods. Unlike traditional models that rely on fixed thresholds, ML approaches allow for dynamic adjustments to individual patient characteristics, leading to more personalized and precise predictions. By incorporating a broader range of clinical features, such as age, sex, and biomarkers, ML models can reduce the risk of misclassification, ensuring that CKD is accurately detected and staged promptly. This study underscores the transformative potential of ML in improving the prediction, diagnosis, and management of CKD. By combining traditional biomarkers with advanced computational methods, clinicians can achieve more accurate, individualized assessments, ultimately enhancing patient care.

This study has several limitations. The lack of direct GFR measurements prevents accurate comparison of eGFR values from equations, limiting precision. The absence of confirmatory tests, such as repeated measurements, also introduces ambiguity in assessing CKD prevalence. While using standardized SCr and SCysC measurements for eGFR calculation is a strength, it relies on indirect estimation, which may lead to inaccuracies. Variations in demographics and comorbidities across populations also limit the generalizability of our findings.

V. CONCLUSION

In conclusion, this study presents an optimized ML-driven framework for CKD classification and staging, demonstrating superior accuracy, robustness, and interpretability compared to traditional methods. By leveraging SHAP for explainability, the model aligns with clinical requirements for transparency and trust, making it a valuable tool for AI-driven precision nephrology. The proposed approach offers a scalable CKD diagnosis and management solution,

potentially enhancing early detection and clinical decision-making. Future research should further validate the developed models using more extensive and diverse datasets to assess their clinical utility and real-world impact. Additionally, incorporating a broader range of clinical features, including biomarkers, lifestyle factors, and genetic data, could improve CKD risk prediction and staging accuracy. Addressing these challenges will enhance the model's reliability, facilitate integration into clinical practice, and improve patient outcomes.

DATA AVAILABILITY

The research described in this study utilized the UK Biobank resource, facilitated under application number 106155. However, the authors cannot publicly share the datasets generated and/or analyzed during this study due to their sensitive nature, potentially compromising research participants' privacy. Nevertheless, interested parties may request access to the data from the corresponding author.

COMPETING INTERESTS

The authors declare that none of their competing financial interests or relationships could influence the findings presented in this work.

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